

< prerequisite >

⌘ Notation

$z_0 \sim p_0$: Source space image

$z_1 = \phi(z_0)$: Target space image

⌘ Jacobian : used to represent the non-linear transformation using linear transformation in 'local' space
(changing the coordinates.)

$$(u, v) \rightarrow (x, y) \quad \left\{ \begin{array}{l} dx = \frac{dx}{du} du + \frac{dx}{dv} dv \\ dy = \frac{dy}{du} du + \frac{dy}{dv} dv \end{array} \right\}$$

$$\Rightarrow \begin{bmatrix} dx \\ dy \end{bmatrix} = \begin{bmatrix} \frac{dx}{du} & \frac{dx}{dv} \\ \frac{dy}{du} & \frac{dy}{dv} \end{bmatrix} \begin{bmatrix} du \\ dv \end{bmatrix}$$

Jacobian

< Normalizer flow >

We should see the whole process in the aspect of 'changing the coordinates.'

$$1. \quad p_1(z_1) dz_1 = p_0(z_0) dz_0$$

$\Rightarrow$ probability mass should be same

$$2. \quad dz_0 = \left| \det \frac{d\phi^{-1}(z_1)}{dz_1} \right| dz_1 \quad (\phi(z_0) = z_1)$$

$\Rightarrow$ Adjusting volume

$$\begin{aligned} \Rightarrow p_1(z_1) &= p_0(z_0) \frac{dz_0}{dz_1} = p_0(\phi^{-1}(z_1)) \left| \det \frac{d\phi^{-1}(z_1)}{dz_1} \right| \\ &= [\phi]_* p_0(z_0) \quad ; \text{ push forward operator } [\phi]_* \end{aligned}$$

$$\text{in Normalizing flow, } \mathcal{L} = -\mathbb{E}_{z_1 \sim p_1} [\log p_1(z_1)] = -\mathbb{E}_{z_1 \sim p_1} [p_1(z_1)]$$

$\Rightarrow$ Maximizing the probability of real image z_1

Since ϕ is invertible, how to find ϕ and its inverse?

$$\rightarrow \text{Residual flow} : \phi = \phi_{(T-1)\Delta t} \circ \dots \circ \phi_{1\Delta t} \circ \phi_{0\Delta t}$$

$$, \quad \phi_t(z_t) = z_t + \Delta t \cdot \underbrace{u_t(z_t)}_{\text{residual update}}$$

Since $\phi_t(x_t) = x_t + \Delta t \cdot u_t(x_t)$,

$$\lim_{\Delta t \rightarrow 0} \frac{\phi_t(x_t) - x_t}{\Delta t} = \frac{x_{t+\Delta t} - x_t}{\Delta t} = \frac{dx_t}{dt} = u_t(x_t)$$

$$\xrightarrow{\text{ODE}} \left[\frac{dx_t}{dt} = u_t(x_t) \right] \implies x_t = x_0 + \int_0^t u_s(x_s) ds$$

$$= x_0 + \int_0^t u_s(\phi_s(x_0)) ds$$

$\hookrightarrow \phi_s = \phi_s(\phi_{s-1}(\dots(\phi_0(\cdot))\dots))$

< Fokker planck & CNF >

SDE is targeting a single data x , but Fokker planck eq explains whole probability.

$$dx_t = f(x_t, t) dt + g(x_t, t) d\omega$$

$$\Rightarrow \frac{\partial p_t(x)}{\partial t} = \underbrace{-\nabla \cdot [f(x_t, t) p_t(x_t)]}_{\text{drift term}} + \underbrace{\frac{1}{2} \nabla^2 [g(x_t, t) g(x_t, t)^T p_t(x_t)]}_{\text{diffusion term}}$$

if $\frac{dx_t}{dt} = u_t(x_t)$ & ODE situation?

$$\Rightarrow \frac{\partial p_t(x_t)}{\partial t} = -\nabla \cdot [u_t(x_t) p_t(x_t)] +$$

$\nearrow \text{drift = ODE}$

$$\Rightarrow \frac{\partial \log p_t(x_t)}{\partial t} = -\nabla \cdot u_t(x_t)$$

Therefore, $\log p_t(x_t) = \log p_0(x_0) + \int_0^t -\nabla \cdot u_s(x_s) ds$

Continuous Normalizing flow : $\mathcal{L} = -\mathbb{E}_{x_1 \sim p_1} [\log p_t(x_1)]$

by learning vector field $u_0(x_0)$

the meaning of divergence ∇

$$\nabla \cdot u(x) = \text{Trace}(\text{Jacobian } u(x)) = \text{Trace}\left(\frac{\partial u}{\partial x}\right)$$

$\hookrightarrow$ 모든 방향 벡터를 $\hookrightarrow$ 각 좌표축에 대해 얼마나 늘어나고 줄어드는가!

< Conditional Flow Matching >

$$\log p_1(x_1) = \log p_0(x_0) + \int_0^1 -\nabla \cdot u_s(x_s) ds$$

$\Rightarrow$ let's learn $u_s(x_s)$ using neural network $V_\theta(z_s)$

but, no information about $u_s(x_s)$.. Then, how??

$$\text{Conditional probability : } p_{\pi_1}(x_0) = \int p_{\pi_1}(x_0|x_1) p_1(x_1) dx_1$$

$$\text{define : } \begin{cases} p_{0|1}(x_0|x_1) = p_0(x_0) = \mathcal{N}(0, I) \\ p_1(x_1) = \mathcal{N}(x_1, \sigma_1^2 I) \end{cases}$$

from Fokker-Planck eq...

$$\frac{\partial p_{\pi_1}(x_0|x_1)}{\partial t} = -\nabla \cdot [u_t(x_0|x_1) p_{\pi_1}(x_0|x_1)]$$

$\Rightarrow$ Conditional vector field $u_t(x_0|x_1)$ is NOT UNIQUE!

$\Rightarrow$ So let's assume ...

$$1. p_{\pi_1}(x_0|x_1) = [\phi_0]_{\#} p_0(x_0)$$

$$2. x_0 \sim p_0, \quad x_t = \phi_t(x_0|x_1)$$

$$\hookrightarrow \text{push forward, recall } p_1(x_1) = p_0(x_0) \frac{dx_0}{dx_1} = p_0(\phi^{-1}(x_1)) \left| \det \frac{\partial \phi^{-1}}{\partial x_1} \right| \\ = [\phi^{-1}]_{\#} p_0(x_0) \quad : \text{push forward operator } [\phi]_{\#}$$

* Path, flow map, vector field

$$1. p_{\pi_1}(x_0|x_1) = \mathcal{N}(\mu_t(x_1), \sigma_t^2(x_1)I)$$

$\rightarrow$ probability path.

$\rightarrow$ Just like forward process in DDPM.

$$2. x_t = \phi(x_0|x_1) = \mu_t(x_1) + \sigma_t(x_1) z_0, \quad z_0 \sim \mathcal{N}(0, I)$$

$\rightarrow$ flow map.

$$3. \quad U_t(x_t | x_1)$$

$$= \frac{d}{dt} x_{t+1}$$

$$= \frac{d}{dt} \phi_t(x_t | x_1)$$

$$= \frac{d}{dt} (\mu_t(x_t) + \sigma_t(x_t) z_t)$$

$$= \mu'_t(x_t) + \sigma'_t(x_t) z_t$$

$$= \frac{\sigma'_t(x_t)}{\sigma_t(x_t)} (x_t - \mu_t(x_t)) + \mu'_t(x_t)$$

→ vector field of x at specific timestep t

$$x_t = \phi(z_t | x_1) = \mu_t(x_1) + \sigma_t(x_1) z_t$$

$$z_t = \frac{1}{\sigma_t(x_t)} (x_t - \mu_t(x_t))$$

So, how does it actually work??

all formulas are conditioned by x_1 , which is not given in inference time!

① we can express marginal vector field ' $U_t(x_t)$ ' with conditional vector field ' $U_t(x_t | x_1)$ '

$$U_t(x_t) = \int U_t(x_t | x_1) \frac{p_{t1}(x_t | x_1) p_1(x_1)}{p_t(x_t)} dx_1$$

proof → in factor graph eq.

$$\frac{d p_{t1}(x_t | x_1)}{dt} = -\nabla \cdot [U_t(x_t | x_1) p_{t1}(x_t | x_1)]$$

$$\frac{d p_t(x_t)}{dt} = \frac{d}{dt} \int p_{t1}(x_t | x_1) p_1(x_1) dx_1$$

$$= - \int \nabla_{x_t} [U_t(x_t | x_1) p_{t1}(x_t | x_1)] p_1(x_1) dx_1$$

$$= - \nabla_{x_t} \int U_t(x_t | x_1) p_{t1}(x_t | x_1) p_1(x_1) \cdot \frac{p_t(x_t)}{p_t(x_t)} dx_1$$

$$= - \nabla_{x_t} \left(\int U_t(x_t | x_1) \frac{p_{t1}(x_t | x_1) p_1(x_1)}{p_t(x_t)} dx_1 \cdot p_t(x_t) \right)$$

$$= - \nabla (U_t(x_t) p_t(x_t)) \rightarrow \text{by factor graph.}$$

$$\text{therefore, } U_t(x_t) = \int U_t(x_t | x_1) \frac{p_{t1}(x_t | x_1) p_1(x_1)}{p_t(x_t)} dx_1$$

② Initial loss $E_{\tau \sim U[0,1]} E_{x_c \sim p_c} [\|V_\theta(x_c, \tau) - u_c(x_c)\|^2]$

is equal to $E_{\tau \sim U[0,1]} E_{x_c \sim p_{c1}} [\|V_\theta(x_c, \tau) - u_{c1}(x_c|x_c)\|^2]$

$$\begin{aligned}
 \xrightarrow{\text{proof}} E_{x_c \sim p_c} \langle V_\theta(x_c, \tau) \cdot u_c(x_c) \rangle &= \int \langle V_\theta(x_c, \tau) \cdot u_c(x_c) \rangle p_c(x_c) \cdot dx_c \\
 &= \int \langle V_\theta(x_c, \tau) \cdot \int u_c(x_c|x_1) \frac{p_{c1}(x_c|x_1) p_c(x_1)}{p_c(x_c)} dx_1 \rangle p_c(x_c) dx_c \\
 &\rightarrow \text{nice trick} \\
 &= \int \int \langle V_\theta(x_c, \tau) \cdot u_c(x_c|x_1) p_{c1}(x_c|x_1) p_c(x_1) \rangle dx_1 dx_c \\
 &= E_{x_1 \sim p_c(x_1), x_c \sim p_{c1}(x_c|x_1)} \langle V_\theta(x_c, \tau) \cdot u_c(x_c|x_1) \rangle
 \end{aligned}$$

loss $\mathcal{L}_\tau$: $\mathcal{L} = E_{\tau \sim U[0,1]} E_{x_1 \sim p_1, x_c \sim p_{c1}(x_c|x_1)} [\|V_\theta(x_c, \tau) - u_{c1}(x_c|x_1)\|^2]$

note that $x_c = \phi_\tau(x_0|x_1)$

$$\mathcal{L} = E_{\tau \sim U[0,1]} E_{x_1 \sim p_1, x_c \sim p_{c1}(x_c|x_1)} [\|V_\theta(\phi_\tau(x_0|x_1), \tau) - u_{c1}(\phi_\tau(x_0|x_1)|x_1)\|^2]$$